

EduMatch — AI-Driven Thesis & Internship Ecosystem

Smart supervisor & internship matchmaker: end-to-end lifecycle management connecting students, supervisors, companies, and alumni. Designed for universities to reduce administrative friction and improve research outcomes.

Team

Code & Query

Members

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Project Snapshot

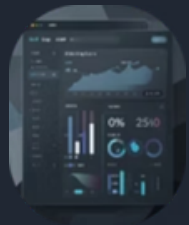
Vision: An AI-powered platform to streamline thesis & internship workflows — matching, tracking, verification, and cross-university collaboration.

Core Problem: Students struggle to find supervisors/internships; supervisors face admin overload; companies lack vetted pipelines; no unified thesis health monitoring.

Target Users: Undergraduate & postgraduate students, faculty supervisors, university admins, corporate recruiters, alumni mentors.



Solution & Key Differentiators



Skill-Gap Analysis

AI identifies gaps between student skills and thesis requirements and recommends targeted free courses.



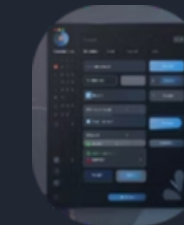
Dual-Blind Matching

Blind profiles with MCDM ranking (CGPA, skills, interest) produce unbiased supervisor-student matches.



Plagiarism & Milestone Tracker

Automated chapter submissions, plagiarism checks, and progress alerts for thesis health monitoring.



Interview Scheduler

Automatic interview slot allocation that syncs with student calendars and company availabilities.

📌 Key outcomes: Faster matching, higher thesis completion rates, trusted credentialing, and an alumni mentorship layer for professional readiness.

Mandatory Project Overview (Concise)

EduMatch is an AI-powered ecosystem that automates thesis and internship lifecycle management for students, supervisors, companies, and alumni. Built to be PHP-friendly and easily mapped to SQL-driven backend logic for fast implementation.

AI Skill-Gap & Course Suggestion

Detect "Missing Skill" and auto-suggest free courses from the database (backend: SQL lookups).

MCDM Matching & Load Balancing

Rank by CGPA, skills, interest. Auto-reassign applicants when supervisor quota is full.

Inter-University Bridge

Cross-institution resource sharing controlled via university_id visibility rules.

Thesis Health & Burnout Detection

Submission pattern analysis flags "At Risk" projects when milestones are late.

Database & ERD Concept (Essential Tables)

Structure designed for clear, SQL-first implementation. Keys and minimal attributes shown for developer handoff.

Users	user_id, name, email, role, university_id
Students	student_id, user_id, cgpa, research_interest, assigned_supervisor_id
Faculty	faculty_id, user_id, designation, quota, current_student_count
Projects/Thesis	project_id, student_id, supervisor_id, status, health_score
Milestones	milestone_id, project_id, due_date, submission_date, plagiarism_score
Alumni_Mentors	alumni_id, user_id, expertise, company
Universities	university_id, uni_name, location

 Design note: All core behaviors can be implemented with concise SQL (e.g., WHERE university_id = X, ORDER BY cgpa DESC, UPDATE quota triggers).

System Architecture & Technical Highlights



Load Balancing: Auto-reassign when faculty quota reached; fallback logic and waitlists.

Concurrency Control: Transactional locks and conflict resolution for simultaneous applications.

MCDM Engine: Composite ranking using CGPA, research interest match, and technical skills.

QR Credentialing: Faculty-verified skills produce QR codes for tamper-proof verification.

Technology Stack & Methodology



Frontend

HTML, responsive UI, calendar sync

Methodology: Waterfall (preferred for clear deliverables and staged university approvals)

Design: Figma prototypes for UI sign-off



Backend

PHP APIs, REST endpoints,
authentication

Implementation note: Keep backend logic SQL-centric for simplicity and maintainability.



Database

MySQL with normalized tables and
indexes for fast matching queries

Project Scope & Core Pages (Reduced feature set)

Per request to simplify, we recommend focusing on a minimal viable product of pages to accelerate development and approvals.

Common

Landing, Login, Registration

Student

Dashboard, Profile & Skills, Matching
Page, Thesis Submission

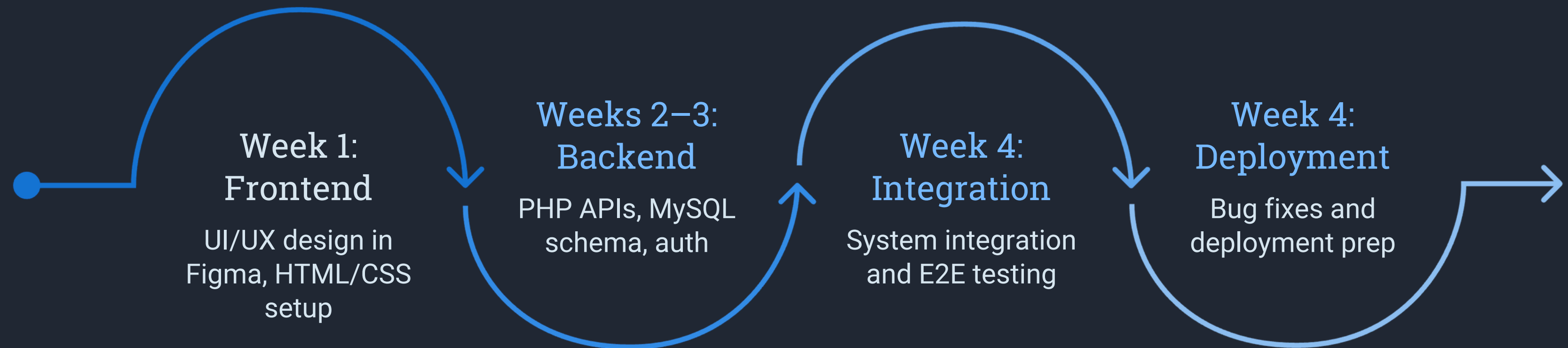
Supervisor

Dashboard, Applicant Review (Blind
Profiles), Capacity Settings

Admin

Main Console, Inter-University Manager, Plagiarism & Health Reports

Timeline (High-level 4-week plan)



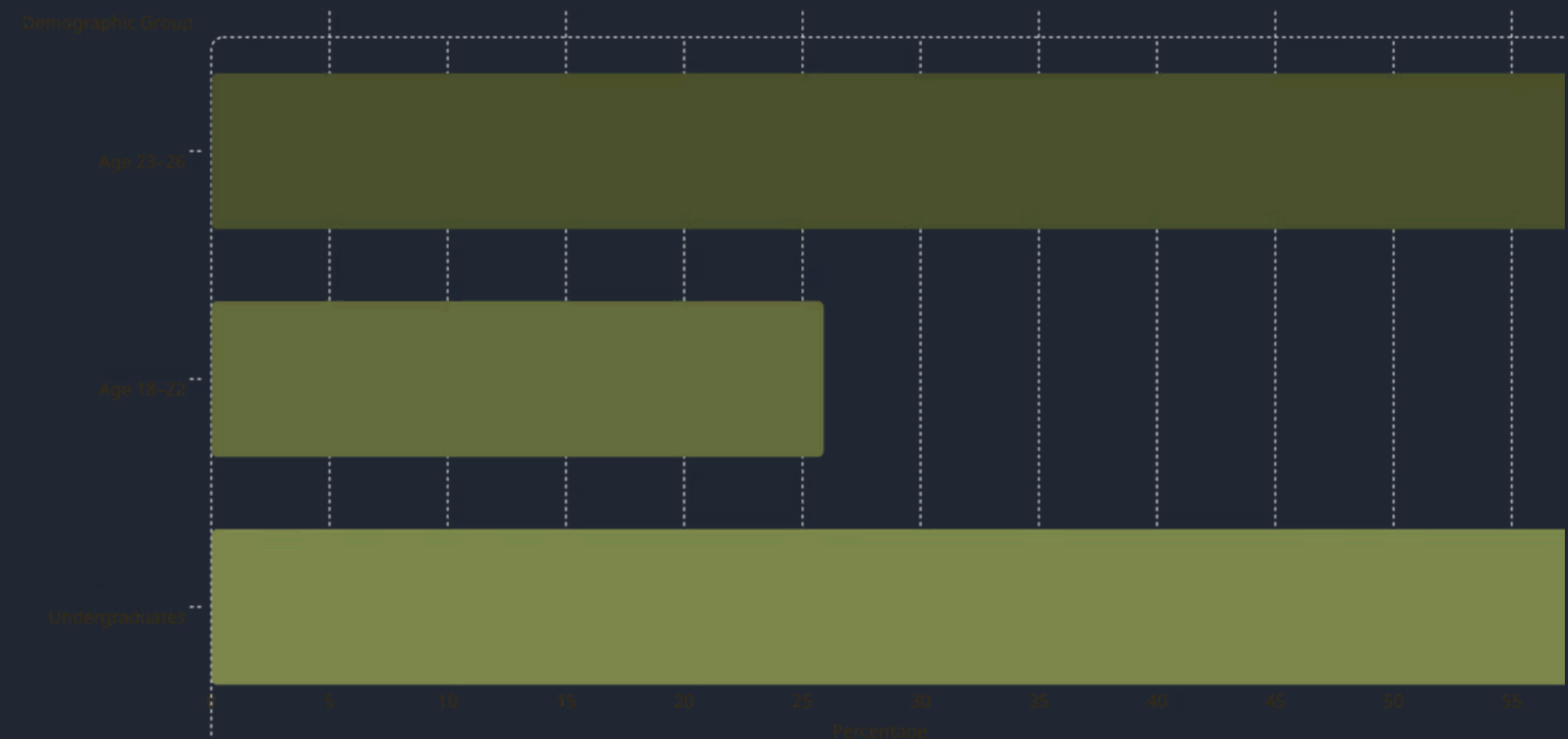
Week 1 — UI/UX design in Figma and HTML/CSS setup. Weeks 2–3 — Backend: PHP APIs, MySQL schema, auth, integration. Week 4 — System integration, E2E testing, bug fixes, deployment preparation.

Research & Next Steps

Survey: n=81 (via Facebook, WhatsApp, university networks). Majority undergraduates from private universities; principal age range 23–26. Insights will guide UX priorities and default matching heuristics.

Immediate next steps:

- Confirm prioritized page list and scope
- Approve data model and SQL query patterns for matching
- Begin Week 1: Figma sign-off and frontend sprint



- ✅ Impact estimate: reduce supervisor-student matching time by ~80% and increase credential trust via QR-backed verification. Share this with the dev team to begin SQL-driven implementation immediately.